

Norm-Attaining Projective Tensors

A dense-envelope theorem, failure of projection descent, and the exact reflexive bottleneck

Research memorandum

June 20, 2026

Abstract

For reflexive Banach spaces X and Y , it remains open whether projective norm-attaining tensors are dense in $X \widehat{\otimes}_\pi Y$. This memorandum does not claim a full solution. It proves, however, a new structural theorem that sharply changes the projection-based approach.

For every pair X, Y we construct a Banach space Y_∞ containing Y isometrically as a 1-complemented subspace such that norm-attaining tensors are dense in $X \widehat{\otimes}_\pi Y_\infty$. Consequently, density of projective norm-attaining tensors does not in general descend to 1-complemented tensor subspaces. Combining the construction with the known non-density example of Dantas, Jung, Roldán, and Rueda Zoca gives a counterexample to the density-descent statement appearing as Lemma 4.1 in a 2025 paper, and therefore also to a 2026 corollary deduced from it, as those statements are printed.

The construction is based on a trace-face correction theorem and a universal graph stabilization: every tensor becomes ambiently approximable by finitely norm-attaining tensors after a single isometric, 1-complemented graph embedding of the second factor. Iterating this embedding produces the dense envelope. Finally, we prove that the original reflexive problem is equivalent to density descent for a very special class of reflexive graph projections. Thus the remaining difficulty is not arbitrary projection theory, which is false, but a precise reflexive common-support correction problem.

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1 The question and notation

All Banach spaces are over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. A tensor $u \in X \widehat{\otimes}_\pi Y$ is said to attain its projective norm if it has a representation

$$u = \sum_{n=1}^{\infty} \lambda_n x_n \otimes y_n, \quad \lambda_n > 0, \quad x_n \in S_X, \quad y_n \in S_Y,$$

with

$$\|u\|_\pi = \sum_{n=1}^{\infty} \lambda_n.$$

We denote the set of such tensors by $\text{NA}_\pi(X \widehat{\otimes}_\pi Y)$. We write $\text{FNA}_\pi(X \widehat{\otimes}_\pi Y)$ for the tensors admitting an optimal representation with only finitely many summands, and put

$$\mathcal{C}_\pi(X, Y) := \overline{\text{FNA}_\pi(X \widehat{\otimes}_\pi Y)}^{\|\cdot\|_\pi}.$$

The source problem is the following.

Question 1.1 (Dantas–Jung–Roldán–Rueda Zoca). *If X and Y are reflexive Banach spaces, is*

$$\overline{\text{NA}_\pi(X \widehat{\otimes}_\pi Y)}^{\|\cdot\|_\pi} = X \widehat{\otimes}_\pi Y?$$

The most recent paper located in the literature search, posted in February 2026, proves that integral projective norm-attaining tensors are approximable by tensors with finite optimal representations, and that classical and integral density are equivalent. It proves a positive result for separable reflexive pairs when one factor has the approximation property, but does not settle [Question 1.1](#) in general [1]. The original question appears in [3, Question 6.1].

we have proved that if H is a complex Hilbert space, every tensor in $H \widehat{\otimes}_\pi H$ attains its projective norm (see Proposition 3.8) and that the set $\text{NA}_\pi(L_p(\mu) \widehat{\otimes}_\pi L_q(\nu))$ is dense in $L_p(\mu) \widehat{\otimes}_\pi L_q(\nu)$ for $1 < p, q < \infty$ and measures μ and ν (see Example 4.12.(b)). However, we do not know what happens in general when both factors are reflexive spaces.

Question 6.1. *Let X, Y be reflexive Banach spaces. Is it true that the set of all norm-attaining tensors is dense in $X \widehat{\otimes}_\pi Y$?*

Let us notice that, by trying to mimic the proof of Theorem 5.1 (and its lemmas) for the nuclear operator case, one would realize that

Figure 1: Source location of Question 6.1 in [3].

2 Idea of the proof

The mechanism is to separate the original reflexive density problem from a false general projection principle. A trace form on $E \widehat{\otimes}_\pi (E^* \oplus_\infty H)$ supplies a canonical face on which finite optimal tensor representations can be forced by a Bishop–Phelps–Bollobás correction. Given a tensor in $X \widehat{\otimes}_\pi Y$, one embeds Y into a graph inside $X^* \oplus_\infty Y$ so that the lifted tensor lies on this trace face. Iterating the graph embedding gives an ambient space Y_∞ containing Y as a 1-complemented subspace and having dense norm-attaining tensors. Since known examples have non-dense norm-attaining tensors in the original pair, density cannot descend through arbitrary 1-complemented subspaces. The remaining reflexive problem is therefore not a general projection-descent problem, but a special graph descent problem where the ambient space is still reflexive.

Lemma 2.1. *For every pair X, Y ,*

$$\overline{\text{NA}_\pi(X \widehat{\otimes}_\pi Y)} = \overline{\text{FNA}_\pi(X \widehat{\otimes}_\pi Y)} = \mathcal{C}_\pi(X, Y).$$

Proof. It is enough to show that every norm-attaining tensor is a norm limit of finitely norm-attaining tensors. Let

$$u = \sum_{n=1}^{\infty} \lambda_n x_n \otimes y_n, \quad \sum_n \lambda_n = \|u\|_{\pi},$$

be optimal. Choose $B \in S_{(X \widehat{\otimes}_{\pi} Y)^*} = S_{\mathcal{L}(X, Y^*)}$ with $\operatorname{Re} B(u) = \|u\|_{\pi}$. Equality in

$$\|u\|_{\pi} = \sum_n \lambda_n \operatorname{Re} B(x_n, y_n) \leq \sum_n \lambda_n$$

forces $\operatorname{Re} B(x_n, y_n) = 1$ for every n . Hence each partial sum is supported by the same B and has projective norm equal to the sum of its coefficients. The partial sums belong to FNA_{π} and converge to u . $\square$

3 A canonical trace face

The basic correction mechanism is a trace face in a tensor product of the form

$$E \widehat{\otimes}_{\pi} (E^* \oplus_{\infty} H).$$

Define the norm-one bilinear form

$$\tau_E(e, (e^*, h)) = e^*(e).$$

We use the scalar Bishop–Phelps–Bollobás theorem in the following standard form: for every $\varepsilon > 0$ there is $\eta > 0$ such that, whenever $e \in S_E$, $e^* \in B_{E^*}$ and $\operatorname{Re} e^*(e) > 1 - \eta$, there are $\tilde{e} \in S_E$ and $\tilde{e}^* \in S_{E^*}$ satisfying

$$\tilde{e}^*(\tilde{e}) = 1, \quad \|e - \tilde{e}\| < \varepsilon, \quad \|e^* - \tilde{e}^*\| < \varepsilon.$$

Theorem 3.1 (Trace-face correction). *Let E, H be Banach spaces and let*

$$G = E^* \oplus_{\infty} H.$$

If $u \in E \widehat{\otimes}_{\pi} G$ satisfies

$$\operatorname{Re} \tau_E(u) = \|u\|_{\pi},$$

then

$$u \in \mathcal{C}_{\pi}(E, G).$$

More precisely, u is approximable by finite tensors having optimal representations supported by τ_E .

Proof. We may assume $\|u\|_{\pi} = 1$. For every $k \in \mathbb{N}$, choose a Bishop–Phelps–Bollobás threshold $\eta_k \in (0, 1)$ corresponding to the error $1/k$. Choose a normalized projective representation

$$u = \sum_{n=1}^{\infty} \lambda_{k,n} e_{k,n} \otimes (f_{k,n}, h_{k,n}),$$

where

$$e_{k,n} \in S_E, \quad (f_{k,n}, h_{k,n}) \in S_G, \quad \sum_n \lambda_{k,n} < 1 + \frac{\eta_k}{k}.$$

Since $\operatorname{Re} \tau_E(u) = 1$,

$$\sum_n \lambda_{k,n} (1 - \operatorname{Re} f_{k,n}(e_{k,n})) < \frac{\eta_k}{k}.$$

Let

$$A_k = \{n : \operatorname{Re} f_{k,n}(e_{k,n}) > 1 - \eta_k\}.$$

Then

$$\sum_{n \notin A_k} \lambda_{k,n} < \frac{1}{k}.$$

For $n \in A_k$, choose $\tilde{e}_{k,n} \in S_E$ and $\tilde{f}_{k,n} \in S_{E^*}$ such that

$$\tilde{f}_{k,n}(\tilde{e}_{k,n}) = 1, \quad \|e_{k,n} - \tilde{e}_{k,n}\| < \frac{1}{k}, \quad \|f_{k,n} - \tilde{f}_{k,n}\| < \frac{1}{k}.$$

Because $\|h_{k,n}\| \leq 1$, we have

$$\|(\tilde{f}_{k,n}, h_{k,n})\| = 1.$$

Set

$$v_k = \sum_{n \in A_k} \lambda_{k,n} \tilde{e}_{k,n} \otimes (\tilde{f}_{k,n}, h_{k,n}).$$

Every atom in this series is normed by τ_E , so

$$\|v_k\|_\pi = \sum_{n \in A_k} \lambda_{k,n};$$

thus v_k is norm-attaining. Moreover,

$$\begin{aligned} \|u - v_k\|_\pi &\leq \sum_{n \notin A_k} \lambda_{k,n} + \sum_{n \in A_k} \lambda_{k,n} (\|e_{k,n} - \tilde{e}_{k,n}\| + \|f_{k,n} - \tilde{f}_{k,n}\|) \\ &< \frac{1}{k} + \frac{2}{k} \left(1 + \frac{\eta_k}{k}\right) \longrightarrow 0. \end{aligned}$$

Finally, the finite partial sums of v_k are finitely norm-attaining, again supported by τ_E . A diagonal choice proves $u \in \mathcal{C}_\pi(E, G)$. $\square$

4 What projections do and do not preserve

Let Z be a 1-complemented subspace of Y , with norm-one projection $P : Y \rightarrow Z$. The induced copy of $X \hat{\otimes}_\pi Z$ in $X \hat{\otimes}_\pi Y$ is isometric because $\operatorname{Id}_X \otimes P$ is a contractive left inverse.

Proposition 4.1 (Correct pointwise descent). *If $u \in X \hat{\otimes}_\pi Z$ and u is norm-attaining when regarded as an element of $X \hat{\otimes}_\pi Y$, then u is norm-attaining in $X \hat{\otimes}_\pi Z$. The analogous assertion holds for finite norm attainment.*

Proof. Take an optimal ambient representation

$$u = \sum_n x_n \otimes y_n, \quad \sum_n \|x_n\| \|y_n\| = \|u\|_{X \hat{\otimes}_\pi Y}.$$

Since $u = (\operatorname{Id}_X \otimes P)u$,

$$u = \sum_n x_n \otimes P y_n$$

in $X \hat{\otimes}_\pi Z$, and

$$\sum_n \|x_n\| \|P y_n\| \leq \sum_n \|x_n\| \|y_n\| = \|u\|_{X \hat{\otimes}_\pi Y} = \|u\|_{X \hat{\otimes}_\pi Z}.$$

The reverse inequality is the definition of the projective norm, so equality holds. Finiteness is preserved if the original representation is finite. $\square$

The condition that the tensor itself lie in the projected range is essential.

Theorem 4.2 (Universal unfolding). *Let $z \in X \widehat{\otimes}_\pi Y$, and choose any representation*

$$z = \sum_{n=1}^{\infty} x_n \otimes y_n, \quad \sum_n \|x_n\| \|y_n\| < \infty.$$

Put

$$v = \sum_{n=1}^{\infty} e_n(x_n) \otimes y_n \in \ell_1(X) \widehat{\otimes}_\pi Y,$$

where $e_n : X \rightarrow \ell_1(X)$ is the n th coordinate embedding. Then v is norm-attaining. The operator

$$P((u_n)_{n \geq 1}) = \left(\sum_{n=1}^{\infty} u_n, 0, 0, \dots \right)$$

is a norm-one projection on $\ell_1(X)$ and

$$(P \otimes \text{Id}_Y)v = e_1(z).$$

Consequently, an arbitrary tensor is the image of a norm-attaining tensor under a norm-one product projection.

Proof. The canonical isometry

$$\ell_1(X) \widehat{\otimes}_\pi Y \cong \ell_1(X \widehat{\otimes}_\pi Y)$$

sends v to $(x_n \otimes y_n)_n$. Therefore

$$\|v\|_\pi = \sum_n \|x_n \otimes y_n\|_\pi = \sum_n \|x_n\| \|y_n\|,$$

which proves norm attainment. The assertions about P follow from

$$\left\| \sum_n u_n \right\| \leq \sum_n \|u_n\|$$

and direct calculation. □

Thus a proof of [Question 1.1](#) cannot rest on a general statement that norm-one projections preserve norm-attaining approximants.

5 Universal graph stabilization

For fixed X, Y , let

$$\Gamma_Y = B_{\mathcal{L}(X, Y^*)}$$

and define

$$\mathcal{U}_X(Y) := Y \oplus_\infty \ell_\infty(\Gamma_Y, X^*).$$

For $B \in \Gamma_Y$, write $B^t : Y \rightarrow X^*$ for

$$(B^t y)(x) = B(x)(y).$$

Define

$$j_Y : Y \longrightarrow \mathcal{U}_X(Y), \quad j_Y(y) = (y, (B^t y)_{B \in \Gamma_Y}).$$

Lemma 5.1. *The map j_Y is an isometry, and its range is 1-complemented in $\mathcal{U}_X(Y)$.*

Proof. For every $y \in Y$,

$$\sup_{B \in \Gamma_Y} \|B^t y\| \leq \|y\|,$$

so $\|j_Y(y)\| = \|y\|$. The map

$$Q_Y(y, (f_B)_{B \in \Gamma_Y}) = j_Y(y)$$

is a projection onto $j_Y(Y)$, and

$$\|Q_Y(y, (f_B))\| = \|y\| \leq \max \left\{ \|y\|, \sup_B \|f_B\| \right\}.$$

Hence $\|Q_Y\| = 1$. □

Let

$$J_Y = \text{Id}_X \otimes j_Y : X \hat{\otimes}_\pi Y \longrightarrow X \hat{\otimes}_\pi \mathcal{U}_X(Y).$$

Since j_Y has a contractive left inverse, J_Y is an isometry.

Theorem 5.2 (Universal graph stabilization). *For every pair of Banach spaces X, Y ,*

$$J_Y(X \hat{\otimes}_\pi Y) \subseteq \mathcal{C}_\pi(X, \mathcal{U}_X(Y)).$$

Thus every tensor in the entire 1-complemented graph copy becomes ambiently approximable by finitely norm-attaining tensors after one universal embedding, independent of the tensor.

Proof. Let $u \in X \hat{\otimes}_\pi Y$, and assume $u \neq 0$. Choose

$$B_0 \in S_{\mathcal{L}(X, Y^*)}$$

with

$$\text{Re } B_0(u) = \|u\|_\pi.$$

On $X \hat{\otimes}_\pi \mathcal{U}_X(Y)$ define the coordinate trace

$$\tau_{B_0}(x, (y, (f_B)_{B \in \Gamma_Y})) = f_{B_0}(x).$$

After reordering coordinates, $\mathcal{U}_X(Y)$ is isometric to

$$X^* \oplus_\infty H_{B_0}$$

for a suitable Banach space H_{B_0} , and τ_{B_0} is the canonical trace from [Theorem 3.1](#). Moreover,

$$\text{Re } \tau_{B_0}(J_Y u) = \text{Re } B_0(u) = \|u\|_\pi = \|J_Y u\|_\pi.$$

The conclusion follows from [Theorem 3.1](#). □

Corollary 5.3 (Local closure descent fails). *If $\mathcal{C}_\pi(X, Y) \neq X \hat{\otimes}_\pi Y$, then*

$$\mathcal{C}_\pi(X, \mathcal{U}_X(Y)) \cap J_Y(X \hat{\otimes}_\pi Y) \subsetneq J_Y(\mathcal{C}_\pi(X, Y)).$$

In particular, ambient approximability of every tensor in a 1-complemented tensor copy does not imply intrinsic approximability in that copy.

Proof. Choose $u \notin \mathcal{C}_\pi(X, Y)$. By [Theorem 5.2](#), $J_Y u$ belongs to the left-hand side. Since J_Y is an isometric tensor isomorphism onto $X \hat{\otimes}_\pi j_Y(Y)$, it maps $\mathcal{C}_\pi(X, Y)$ onto the intrinsic closure $\mathcal{C}_\pi(X, j_Y(Y))$. Hence $J_Y u$ does not belong to the right-hand side. □

6 A dense norm-attaining envelope

The previous construction can be iterated.

Theorem 6.1 (Dense-envelope theorem). *For every pair of Banach spaces X, Y , there are a Banach space Y_∞ and an isometric embedding*

$$\iota : Y \longrightarrow Y_\infty$$

such that:

- (i) $\iota(Y)$ is 1-complemented in Y_∞ ;
- (ii)

$$\mathcal{C}_\pi(X, Y_\infty) = X \widehat{\otimes}_\pi Y_\infty.$$

Proof. Set $Y_0 = Y$. Inductively let

$$Y_{n+1} = \mathcal{U}_X(Y_n)$$

and embed Y_n into Y_{n+1} by the graph isometry $j_n := j_{Y_n}$. Let

$$p_n : Y_{n+1} \longrightarrow Y_n$$

be the contractive left inverse given by the first coordinate, so that

$$p_n j_n = \text{Id}_{Y_n}.$$

Identify each Y_n with its image in Y_{n+1} and let Y_∞ be the completion of the resulting increasing union.

For $m > n$, put

$$p_{n,m} = p_n p_{n+1} \cdots p_{m-1} : Y_m \rightarrow Y_n.$$

The relations $p_m j_m = \text{Id}$ make these retractions compatible with the identifications. Hence they define a contractive projection from the algebraic union onto Y_n , and this projection extends to Y_∞ . Thus every Y_n , in particular Y_0 , is 1-complemented in Y_∞ .

Now take $u \in X \widehat{\otimes}_\pi Y_n$. Under the identification of Y_n with its graph image in Y_{n+1} , [Theorem 5.2](#) gives

$$u \in \mathcal{C}_\pi(X, Y_{n+1}).$$

Since Y_{n+1} is 1-complemented in Y_∞ , the induced inclusion

$$X \widehat{\otimes}_\pi Y_{n+1} \longrightarrow X \widehat{\otimes}_\pi Y_\infty$$

is isometric. Every finite optimal representation in $X \widehat{\otimes}_\pi Y_{n+1}$ remains optimal in $X \widehat{\otimes}_\pi Y_\infty$. Therefore

$$X \widehat{\otimes}_\pi Y_n \subseteq \mathcal{C}_\pi(X, Y_\infty) \quad (n \in \mathbb{N}).$$

The union of these tensor subspaces is dense in $X \widehat{\otimes}_\pi Y_\infty$, because $\bigcup_n Y_n$ is dense in Y_∞ and algebraic tensors are dense in the projective tensor product. Since $\mathcal{C}_\pi(X, Y_\infty)$ is closed, it is the whole space. $\square$

Corollary 6.2 (Density does not descend to 1-complemented subspaces). *There exist Banach spaces X, Y, Z such that Y is isometric to a 1-complemented subspace of Z ,*

$$\mathcal{C}_\pi(X, Z) = X \widehat{\otimes}_\pi Z,$$

but

$$\mathcal{C}_\pi(X, Y) \neq X \widehat{\otimes}_\pi Y.$$

Proof. Dantas, Jung, Roldán, and Rueda Zoca constructed Banach spaces X, Y for which norm-attaining tensors are not dense in $X \widehat{\otimes}_\pi Y$ [[3](#), Theorem 5.1]. Apply [Theorem 6.1](#) to that pair and take $Z = Y_\infty$. $\square$

6.1 Consequence for two printed descent statements

Lemma 4.1 of [4] states that, if norm-attaining tensors are dense in $X \widehat{\otimes}_\pi Y$ and $Z \subseteq Y$ is 1-complemented, then norm-attaining tensors are dense in $X \widehat{\otimes}_\pi Z$. Its proof chooses an ambient norm-attaining approximant z' and asserts that $(\text{Id} \otimes P)z'$ is norm-attaining “as in Lemma 3.1.” Lemma 3.1 applies when the tensor being represented already lies in the projected range, as in Proposition 4.1; it does not apply to an arbitrary approximant z' . The universal unfolding theorem shows directly that projection can destroy norm attainment, and Corollary 6.2 gives a counterexample to the density statement itself.

Corollary 6.3. *The density-descent assertion in [4, Lemma 4.1], as printed, is false in general.*

The 2026 integral-representation preprint deduces its Corollary 2.14 from that lemma and from the equivalence between classical and integral density [1, Corollaries 2.12 and 2.14]. Therefore the same dense-envelope example also contradicts Corollary 2.14 as printed: integral norm-attaining tensors are dense in the envelope, but not in the complemented bad subspace.

This observation does not affect results whose proofs do not use the faulty density descent. It does mean that downstream applications relying on that step require re-examination.

7 A separable version

The universal envelope uses a large ℓ_∞ -sum. When X^* and Y are separable, a diagonal construction produces a separable envelope.

Theorem 7.1 (Separable dense envelope). *Assume that X^* and Y are separable. Then the space Y_∞ in Theorem 6.1 may be chosen separable.*

Proof. We construct separable spaces Y_n inductively. Start with $Y_0 = Y$. Once Y_n has been constructed, choose a countable dense set

$$D_n = \{u_{n,1}, u_{n,2}, \dots\} \subset X \widehat{\otimes}_\pi Y_n.$$

Such a set exists because separability of X^* implies separability of X , and hence $X \widehat{\otimes}_\pi Y_n$ is separable.

At stage n , view the finitely many tensors

$$u_{k,j}, \quad 0 \leq k \leq n, \quad 1 \leq j \leq n,$$

in $X \widehat{\otimes}_\pi Y_n$ through the preceding embeddings. For every nonzero tensor v in this finite set choose a support

$$B_v \in S_{\mathcal{L}(X, Y_n^*)}, \quad \text{Re } B_v(v) = \|v\|_\pi.$$

Let F_n be the resulting finite family and put

$$Y_{n+1} = Y_n \oplus_\infty (X^*)_{\ell_\infty}^{F_n}.$$

Embed Y_n by

$$y \longmapsto (y, (B_v^\dagger y)_{v \in F_n}).$$

This is an isometry with a contractive first-coordinate retraction. For each selected v , its image is normed by the coordinate trace belonging to B_v , and therefore belongs to $\mathcal{C}_\pi(X, Y_{n+1})$ by Theorem 3.1.

Take the completion of the increasing union. Every $u_{k,j}$ is selected at some later stage, so its final image belongs to $\mathcal{C}_\pi(X, Y_\infty)$. Since D_k is dense in $X \widehat{\otimes}_\pi Y_k$, the entire image of $X \widehat{\otimes}_\pi Y_k$ lies in the closed set $\mathcal{C}_\pi(X, Y_\infty)$. The union over k is dense, proving density in the limit. Every Y_n is separable because only finitely many copies of the separable space X^* are added at each step; hence the limit is separable. $\square$

In the non-density example of [3, Theorem 5.1], the first factor is a subspace of c_0 and therefore has separable dual, while the second tensor factor may also be taken separable. Thus the counterexample in Corollary 6.2 can be realized with a separable ambient space.

8 The exact reflexive bottleneck

The dense-envelope construction is highly nonreflexive. This is not incidental: it shows that reflexivity is doing all the remaining work.

Consider the following property.

Definition 8.1 (Reflexive descent property). We say that *reflexive closure descent* holds if, whenever X, F are reflexive and $Z \subseteq F$ is a reflexive 1-complemented subspace,

$$\mathcal{C}_\pi(X, F) \cap (X \widehat{\otimes}_\pi Z) = \mathcal{C}_\pi(X, Z),$$

where $X \widehat{\otimes}_\pi Z$ is identified with its canonical isometric copy in $X \widehat{\otimes}_\pi F$.

Theorem 8.2 (Equivalence with the original problem). *The following are equivalent.*

- (i) *For every pair of reflexive Banach spaces X, Y ,*

$$\mathcal{C}_\pi(X, Y) = X \widehat{\otimes}_\pi Y.$$

- (ii) *Reflexive closure descent from Definition 8.1 holds.*

- (iii) *Closure descent holds for the following special graph embeddings: whenever X, Y are reflexive, $a \in X \widehat{\otimes}_\pi Y$, and*

$$B \in S_{\mathcal{L}(X, Y^*)}, \quad \operatorname{Re} B(a) = \|a\|_\pi,$$

the graph map

$$j_B : Y \rightarrow X^* \oplus_\infty Y, \quad j_B(y) = (B^t y, y),$$

satisfies

$$(\operatorname{Id}_X \otimes j_B)a \in \mathcal{C}_\pi(X, j_B(Y))$$

when the graph range is considered intrinsically.

Proof. The implication (i) $\Rightarrow$ (ii) is immediate: all three tensor spaces in question have full closure.

The implication (ii) $\Rightarrow$ (iii) follows because $X^* \oplus_\infty Y$ is reflexive and $j_B(Y)$ is 1-complemented. Indeed, the projection

$$Q_B(f, y) = (B^t y, y)$$

has norm one. The trace form

$$\tau_X(x, (f, y)) = f(x)$$

satisfies

$$\operatorname{Re} \tau_X((\operatorname{Id}_X \otimes j_B)a) = \operatorname{Re} B(a) = \|a\|_\pi.$$

By Theorem 3.1, the graph image belongs to

$$\mathcal{C}_\pi(X, X^* \oplus_\infty Y).$$

Reflexive descent then places it in the intrinsic closure of the graph range.

Finally, (iii) $\Rightarrow$ (i): for arbitrary $a \in X \widehat{\otimes}_\pi Y$, choose a supporting B as above. The map $\operatorname{Id}_X \otimes j_B$ is an isometric tensor isomorphism from $X \widehat{\otimes}_\pi Y$ onto $X \widehat{\otimes}_\pi j_B(Y)$ and carries finitely norm-attaining tensors to finitely norm-attaining tensors in both directions. Hence intrinsic approximability of the graph image implies $a \in \mathcal{C}_\pi(X, Y)$. $\square$

Thus Question 1.1 is exactly a very special reflexive graph-descent problem. Arbitrary descent is false by Corollary 6.2; the open issue is whether the extra reflexive geometry prevents the failure.

9 A fixed-support correction formulation

The graph proof also gives a concrete sequence of almost maximizing atoms.

Proposition 9.1 (Weighted fixed-support approximation). *Let $a \in S_{X \widehat{\otimes}_\pi Y}$ and let*

$$B \in S_{\mathcal{L}(X, Y^*)}, \quad \operatorname{Re} B(a) = 1.$$

There are finite tensors

$$w_n = \sum_{j=1}^{m_n} \lambda_{n,j} x_{n,j} \otimes y_{n,j} \longrightarrow a$$

with $\lambda_{n,j} > 0$, $x_{n,j} \in S_X$, $y_{n,j} \in B_Y$, such that

$$\sup_n \sum_j \lambda_{n,j} < \infty$$

and

$$\sum_{j=1}^{m_n} \lambda_{n,j} (1 - \operatorname{Re} B(x_{n,j}, y_{n,j})) \longrightarrow 0.$$

Moreover, the atoms arise as the second-coordinate projections of finite optimal tensors in the trace face of

$$X \widehat{\otimes}_\pi (X^* \oplus_\infty Y).$$

Proof. Let

$$j_B(y) = (B^t y, y), \quad \tilde{a} = (\operatorname{Id}_X \otimes j_B) a.$$

By [Theorem 3.1](#), choose finite optimal tensors

$$u_n = \sum_{j=1}^{m_n} \lambda_{n,j} x_{n,j} \otimes (f_{n,j}, y_{n,j}) \longrightarrow \tilde{a}$$

with

$$f_{n,j}(x_{n,j}) = 1, \quad x_{n,j} \in S_X, \quad \|(f_{n,j}, y_{n,j})\| = 1,$$

and

$$\sum_j \lambda_{n,j} = \|u_n\|_\pi.$$

Projecting onto the Y coordinate gives $w_n \rightarrow a$. Also,

$$\begin{aligned} \sum_j \lambda_{n,j} (1 - \operatorname{Re} B(x_{n,j}, y_{n,j})) &= \tau_X(u_n) - \operatorname{Re} B(w_n) \\ &\longrightarrow \tau_X(\tilde{a}) - \operatorname{Re} B(a) = 0. \end{aligned}$$

The coefficient sums converge to $\|\tilde{a}\|_\pi = 1$. □

A full positive answer would follow from a simultaneous correction principle converting the atoms in [Proposition 9.1](#) into exact maximizers of one common nearby bilinear form while changing their weighted barycentre by $o(1)$. Ordinary Bishop–Phelps–Bollobás correction is insufficient because it permits the supporting functional to vary from atom to atom. The common-support constraint is precisely what makes a corrected finite representation optimal.

This is closely analogous to the approximate hyperplane series philosophy, but now for elementary tensors and a common bilinear support. Proving such a principle for reflexive pairs would settle [Question 1.1](#); constructing a reflexive supported tensor for which it fails uniformly would yield a counterexample.

10 Conclusions

The original reflexive question remains unresolved in this memorandum. The additional push nevertheless produced three concrete advances.

First, the universal graph stabilization and its iteration give the dense-envelope theorem: every tensor pair embeds, with the second factor 1-complemented, into a pair having dense norm-attaining tensors. Second, this construction proves that density does not descend through arbitrary 1-complemented subspaces and identifies a gap in a 2025 printed proof and a dependent 2026 corollary. Third, the reflexive problem is now equivalent to descent for a sharply prescribed family of reflexive graph projections, or equivalently to a fixed-support simultaneous correction principle.

The next decisive step must use reflexivity in a way unavailable to the dense-envelope construction. The most promising targets are weak compactness of the graph maximizing face, fragmentation or dentability of the associated barycentric measures, and a common-support version of the approximate hyperplane series property. None of these has yet been proved in sufficient generality, and no reflexive counterexample was found.

References

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